

Nithin Govindarajan

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Areas of expertise

Numerical software development, numerical linear algebra, numerical analysis, PDE solvers, dynamical systems and control, signal processing, computational algebraic geometry, tensor methods.

Education

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|-----------------------|---|
| Oct. 2014 - Dec. 2018 | PHD in Mechanical Engineering, University of California in Santa Barbara
<i>Dissertation title:</i> “Periodic approximations and spectral analysis of the Koopman operator: theory and applications”.
<i>Advisors:</i> I. Mezić, S. Chandrasekaran. |
| Sep. 2009 - Oct. 2012 | MSC in Aerospace Engineering (with distinction), Technische Universiteit Delft
<i>Master thesis:</i> “An Optimal Control Approach for Estimating Aircraft Command Margins”.
<i>Advisors:</i> Q.P. Chu, C.C. de Visser. |
| Sep. 2006 - Jul. 2009 | BSC in Aerospace Engineering (with distinction), Technische Universiteit Delft |

Work experience

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| Nov. 2023 - present | Founder, VGN Computational consultancy services: numerical algorithms, scientific computing, and research software development. |
| Nov. 2018 - present | Co-founder and technical lead, www.matisse.ai : development of production software for AI-assisted dental applications. |
| Aug. 2019 - Feb. 2026 | Postdoctoral researcher, KU Leuven, Belgium |
| Feb. 2019 - July 2019 | Lecturer Mathematics, University of Amsterdam (UvA), The Netherlands |
| Nov. 2012 - May 2013 | Junior R&D engineer, National Aerospace Laboratory (NLR), Amsterdam, The Netherlands |
| Sep. 2011 - May 2012 | Intern, Mission Critical Technologies Inc. (on site at NASA Ames), Moffet Field, CA |

Selected software and numerical development

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| 2022 - present | One of the main developers of PyTensorlab , a Python package for tensor computations: implemented numerical routines, research prototypes, and user-facing scientific software. pytensorlab.net |
| 2024 - present | Developed SSSMatrices in Julia for sequentially semi-separable matrix representations, including conversion from dense matrices, matrix arithmetic, and linear-system solvers. GitHub repository |
| 2022 - present | Developed and validated fast algorithms for Macaulay nullspaces and structured linear algebra, with reproducible code accompanying SIAM publications and technical reports. |

2014 - present Research software for numerical PDEs, dynamical systems, tensor methods, and low-rank structured matrix algorithms in Matlab, Python, and Julia.

Fellowships, honors & awards

Apr. 2016 CCDC fellowship, Center for Control, Dynamics and Computation, Santa Barbara, CA
 Apr. 2014 Department Merit Fellowship (UCSB Mech. Eng), Santa Barbara, CA
 May 2011 Huygens Scholarship Programme, Nuffic, The Hague, The Netherlands

Publications

Journal Publications

2025 Govindarajan, N., Chandrasekaran, S., Dewilde Lathauwer, L. (2025). Tree Quasi-separable Matrices: A Simultaneous Generalization of Sequentially and Hierarchically Semiseparable Representations. *SIAM Journal on Matrix Analysis and Applications*, 46(2), 1562-1586.

2024 Govindarajan, N., Widdershoven, R., Chandrasekaran, S., & De Lathauwer, L. (2023). A fast algorithm for computing Macaulay nullspaces of bivariate polynomial systems. *SIAM Journal on Matrix Analysis and Applications*, 45(1), 368-396.

2022 Govindarajan, N., Vervliet, N., & De Lathauwer, L. (2022). Regression and classification with spline-based separable expansions. *Frontiers in big Data*, 5, 688496.

2022 Govindarajan, N., Epperly, E. N., & Lathauwer, L. D. (2022). $(L_r, L_r, 1)$ -Decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials. *SIAM Journal on Matrix Analysis and Applications*, 43(2), 912-938.

2021 Epperly, E. N., Govindarajan, N., & Chandrasekaran, S. (2021). Minimal rank completions for overlapping blocks. *Linear Algebra and its Applications*, 627, 185-198.

2021 Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2021). On the approximation of Koopman spectra of measure-preserving flows. *SIAM Journal on Applied Dynamical Systems*, 20(1), 232-261.

2019 Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2019). On the approximation of Koopman spectra for measure preserving transformations. *SIAM Journal on Applied Dynamical Systems*, 18(3), 1454-1497.

2015 Govindarajan, N., De Visser, C. C., Van Kampen, E., Krishnakumar, K., Barlow, J., & Stepanyan, V. (2015). Optimal control framework for estimating autopilot safety margins. *Journal of Guidance, Control, and Dynamics*, 38(7), 1197-1207.

2014 Govindarajan, N., de Visser, C. C., & Krishnakumar, K. (2014). A sparse collocation method for solving time-dependent HJB equations using multivariate B-splines. *Automatica*, 50(9), 2234-2244.

Conference proceedings

2023 Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2023, September). Overdetermined systems of polynomial equations: tensor-based solution and application. *Proceedings of EU-SIPCO 2023*, Helsinki, Finland.

2018 Chandrasekaran, S., Govindarajan, N., & Rajagopal, A. (2018, July). Fast Algorithms for Displacement and Low-Rank Structured Matrices. In *Proceedings of the 2018 ACM International Symposium on Symbolic and Algebraic Computation* (pp. 17-22).

2016 Govindarajan, N., Arbabi, H., Van Blargian, L., Matchen, T., & Tegling, E. (2016, December). An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum. In *2016 IEEE 55th Conference on Decision and Control (CDC)* (pp. 6477-6484). IEEE.

Preprints & Tech reports

- 2026 Govindarajan, N., Vervliet, N., Widdershoven, R., De Lathauwer, L. (2026). Efficient Krylov methods for overdetermined polynomial systems. Technical Report 26-108, ESAT-STADIUS, KU Leuven, Leuven, Belgium.
- 2025 Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2025). Fast Macaulay null space through the intersection of shifted null spaces. Technical Report 25-56, ESAT-STADIUS, KU Leuven.
- 2019 Chandrasekaran, S., Epperly, E. N., Govindarajan, N. (2019). Graph-induced rank structures and their representations. arXiv preprint arXiv:1911.05858.

Selected talks

- 2024 “Macaulay matrices, low displacement rank, and the efficient computation of null spaces”, Back to the roots seminar, Leuven.
- 2023 “A tensor-based approach to solving systems of multivariate polynomials”, CAM23, Selva di Fasano.
- 2023 “Efficient Computation of Macaulay Matrix Null Spaces Through Exploiting Shift-Invariant Structures”, SIAM AG23, Eindhoven.
- 2021 “ $(L_r, L_r, 1)$ -decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials”, SeLMA meeting, Leuven.
- 2018 “Spline-based separable expansions for approximation, regression and classification”, IPAM Workshop I: Tensor Methods and their Applications in the Physical and Data Sciences, UCLA (online)
- 2017 “A toolbox for computing spectral properties of dynamical systems”, SIAM DS17, Snowbird.
- 2016 “An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum”, IEEE CDC 16, Las Vegas.

Teaching

Lead instructor & course organizer

- Semester 1 2019/2020 Numerical mathematics, Amsterdam University College

Co-instructor

- March 2022 Fast algorithms for dense structured matrices, KU Leuven
- Semester 2 2021/2022 Numerieke modellering & benadering, KU Leuven
- Semester 2 2019/2020 Numerieke modellering & benadering, KU Leuven

Teaching assistant

- Semester 1 2021/2022 Numerieke wiskunde, KU Leuven
- Semester 1 2020/2021 Numerieke wiskunde, KU Leuven
- Semester 1 2019/2020 Numerieke wiskunde, KU Leuven
- Spring 2018 Control theory, UCSB
- Winter 2017 Electrical circuits Lab, UCSB
- Fall 2017 Intro to programming in Matlab, UCSB
- Summer 2017 Dynamics, UCSB
- Summer 2017 Physics Lab: intro to classical mechanics for non-engineers, UCSB
- Spring 2017 Dynamics, UCSB
- Spring 2015 Dynamics, UCSB

Winter 2015
Fall 2014

Vibrations, UCSB
Statics, UCSB

Extra-curricular

Software skills

Python, Matlab, Julia, Git, LaTeX, Mathematica; working knowledge of C++.

Languages

English, Dutch, Tamil (basic).

Last updated: September 6, 2026